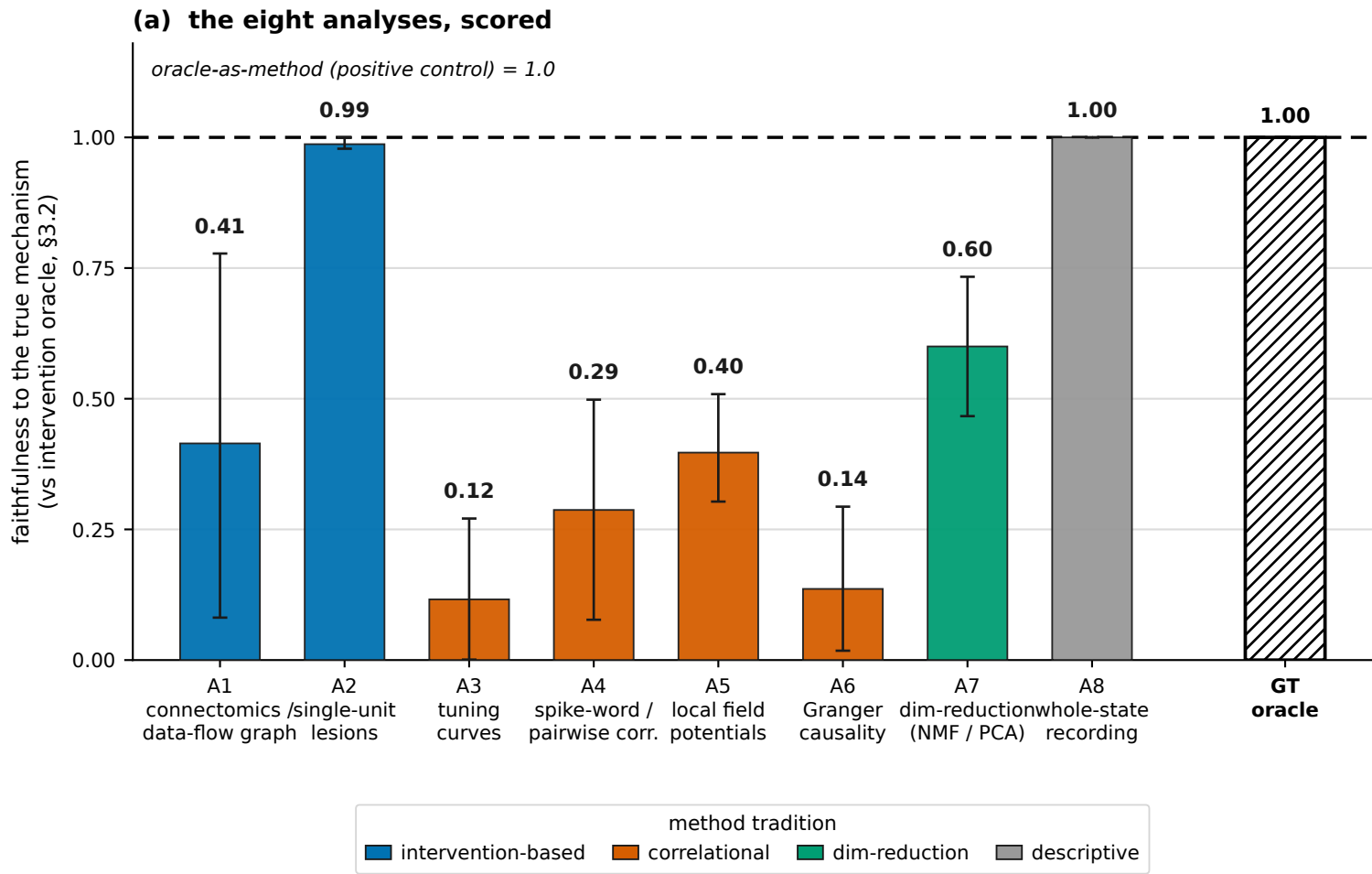


The neuroscience battery on a known machine — scored against ground truth

Jonas & Kording's battery, quantified: classical analyses find rich structure yet score low in faithfulness to the true mechanism.



Data: committed §R records (6 core games); whiskers = 95% bootstrap CI (every bar has a CI record). See Supplement provenance table.

(b) key findings

- A6** Granger F = 0

Pong false-edge 1.00, missed 1.00 → F = 0.00

Temporal precedence ≠ causation: a clock-locked machine yields spurious bidirectional edges.
- A3** spurious tuning

Pong rate 1.00; Space Inv. 0.67; Q*bert 0.53

Strongly-tuned units whose tuning ≠ true causal role — the tuning-curve trap, quantified.
- A2** $\rho = 0.99$, blind to interactions

$\rho = 0.99$ vs true role; misses 40% of interactions

Super-additive pair Δ up to 1044 (Space Inv.) is invisible to one-unit-at-a-time ablation.
- A7** NMF = PCA (both mediocre)

NMF matched frac 0.60 = PCA 0.60 (best 0.8)

Tied on matched components; NMF leads only on the recon-error composite — neither recovers the basis.